

Statistical Decision Making 45-752

Analyzing Data That Vary Over Time

This file is titled: 10_slides.pptx
The associated script is: 10_notebook.ipnyb

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Topics In this Slide Set

[Models with lagged variables](#)

[Analyzing demand for coffee](#)

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Understanding lagged variables

Often in analyzing data that vary over time, we use past values of a variable.
The past value of a variable is called the lagged value of a variable.
To illustrate, we will study daily temperatures in Pittsburgh in May of 2000.
The data are in file PghTempsMay2000.xlsx

Here is a snapshot of the temperatures for the first eight days of the month.

Consider day 5:

- The temperature that day was 70.6.
- The temperature the day before was 63.2.
- The temperature two days before was 57.2.

Hence, on day 5, the one and two period lags of temperature were 63.2 and 57.2.

Day	Temp
1	58.8
2	53.9
3	57.2
4	63.2
5	70.6
6	72.5
7	74.5
8	74.0

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Using Python to Calculate Lagged Variables

Create one-period and two-period lags of Temp

```
# Create one-period and two-period lags of temperature
TempLag1 = Temp.shift(1)
TempLag2 = Temp.shift(2)
```

Put the variables in a spreadsheet and print first 8 rows.

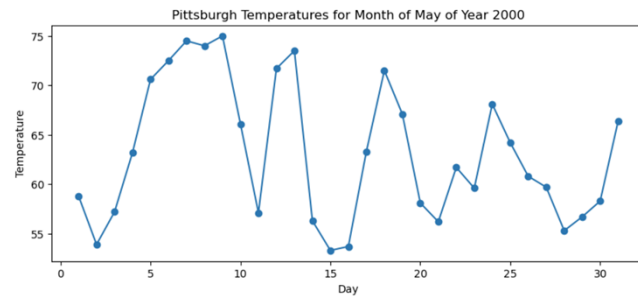
```
# Put Day and the three temperature variables in a data frame
TempKeep = pandas.DataFrame({"Day": Day, "Temp": Temp, "TempLag1": TempLag1, "TempLag2": TempLag2})
# Select and print first 8 rows
first_8_rows = TempKeep.head(8)
print(first_8_rows)
```

Day	Temp	TempLag1	TempLag2
1	58.8	NaN	NaN
2	53.9	58.8	NaN
3	57.2	53.9	58.8
4	63.2	57.2	53.9
5	70.6	63.2	57.2
6	72.5	70.6	63.2
7	74.5	72.5	70.6
8	74.0	74.5	72.5

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View a Plot of The Data



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Estimate Regression Using Lagged Variables

```
TempReg = tprstats.model("ts", "Temp~lag(Temp,1)+lag(Temp,2)", TempsKeep)
TempReg.summary()
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	41.0964	11.217	3.664	0.001	18.039	64.154
lag(Temp, 1)	0.7694	0.179	4.299	0.000	0.402	1.137
lag(Temp, 2)	-0.4126	0.115	-3.572	0.001	-0.650	-0.175

My objective with this example is to help you see how to use the lag() command and how to interpret the results, not to make you a weather forecaster.

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Estimating A Demand Function for Coffee

As an introduction to analyzing time series data, we will estimate a demand function for coffee.

The Coffee Marketers Association would like to know why coffee consumption per capita has declined over time.

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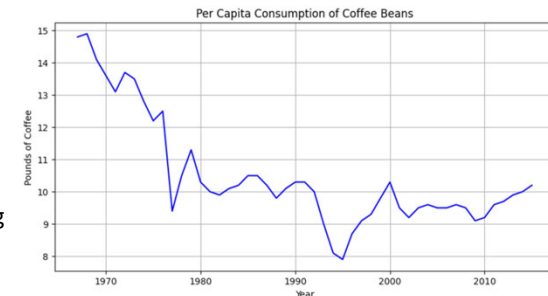
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Coffee Growers' Conundrum: Falling US Consumption

The graph below shows per capita coffee bean consumption in the US

Why did coffee consumption decline?

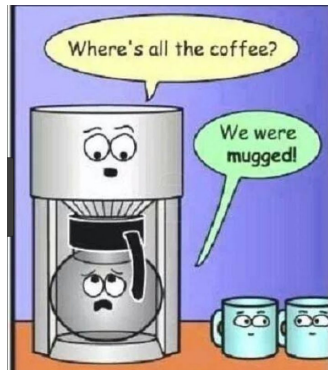
Suppose the Coffee Growers Association hired your consulting firm to investigate.



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One Possible Explanation



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Time Series Data

- Many applications of regression involve use of data that evolve over time.
- We previously estimated a model of demand for natural gas using time series data.
- Regressions with time series often have error terms that are correlated over time.
- When correlation of errors is present, we need to adapt our estimation methods and calculation of standard errors accordingly.
- Heteroskedasticity may also be present with time series data.
- An analysis of coffee demand nicely illustrates these issues.
- For some time series applications, meaningful results require estimating a model using period-to-period changes in the variables.
- We will consider such a situation in a future application of time series analysis.

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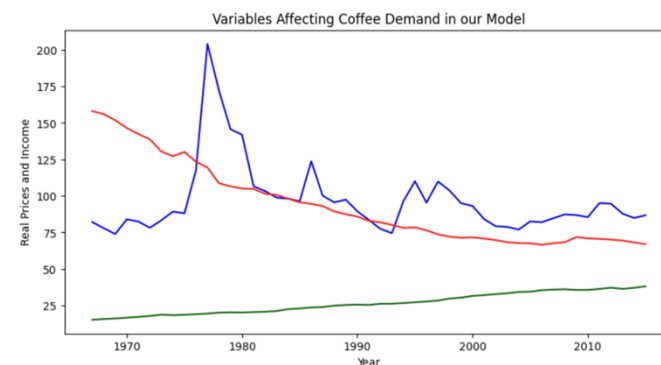
Coffee Demand Function

- Dependent Variable: Coffee Bean Consumption per Capita in the U.S.
- Independent Variables:
 - Per Capita Real Income
 - Real Price of Coffee
 - Real Price of Carbonated Beverages (substitute good)
- The word “Real” means adjusted for inflation.
- Demand functions, supply functions, and cost functions are always estimated with inflation-adjusted data.

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Factors Affecting Demand



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Why The Spike in Prices 1976-1980

The world's coffee trade . . . may have been permanently altered by the frost. Few of the Paraná coffee bushes will recover, and many will not be replaced. The farmers have been frost-bitten too often in the past. They are planning to grow wheat and particularly soya beans.
—The Economist, July 26, 1975

	RPCOFEE
1976	117.3824
1977	204.0985
1978	171.9988
1979	145.6424
1980	141.8552

Brazilian coffee farmers thought they had suffered through every kind of drought or frost, but 1975 brought snow to Paraná for the first time, and the ripples from this freak weather system would affect the global coffee industry for years. Hitting on July 17 and 18, it was by far the worst frost of the century, virtually destroying the Paraná coffee lands, while it inflicted terrible damage in São Paulo and elsewhere.

<https://books.google.com/books?id=TUo98IrkwoC&pg=PA290&lpg=PA290&dq=coffee+freeze+1976+brazil&source=bl&ots=Be4subVYKR&sig=lgTBMWVTqkljgSErMnNQ0e5giHKg&hl=en&sa=X&ei=cblOVKIHqNuKkhNAC&ved=0CEMQ6AEwCA#v=onepage&q=coffee%20freeze%201976%20brazil&f=false>

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Coffee Demand

Our data are in file Coffee_Data.xlsx

Import the data. Then estimate and view the following regression:

```
# Estimate and view coffee regression.
CoffeeReg = tprstats.model("ts", "cons~rpcofee+rpcinc+rpcarb", Coffee_Data)
CoffeeReg.summary()
```

With time series data, we use ts in the regression command.

The result is on the next slide.

Note: The government agency that collected these data discontinued collecting the series in 2016. For our purposes, we have data for a long enough time period that is more than sufficient to investigate factors affecting demand for coffee.

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Coffee Demand

back

We obtain the regression shown below.

```
=====
OLS Regression Results
=====
Dep. Variable:      cons      R-squared:      0.907
Model:              OLS      Adj. R-squared:    0.901
No. Observations:   49      F-statistic:    189.3
Covariance Type:    HAC      Prob (F-statistic): 1.59e-25
=====
              coef      std err      t      P>|t|      [0.025      0.975]
-----
Intercept      1.1211      1.940      0.578      0.566      -2.786      5.028
rpcofee      -0.0132      0.004     -3.469      0.001     -0.021     -0.006
rpcinc       0.1121      0.034      3.279      0.002      0.043      0.181
rpcarb       0.0821      0.009      9.517      0.000      0.065      0.099
=====
```

Are coefficients of the variables significant at the 5% level?

Do the coefficients have the expected algebraic signs?

Do the estimates suggest that carbonated beverages are a substitute for coffee?

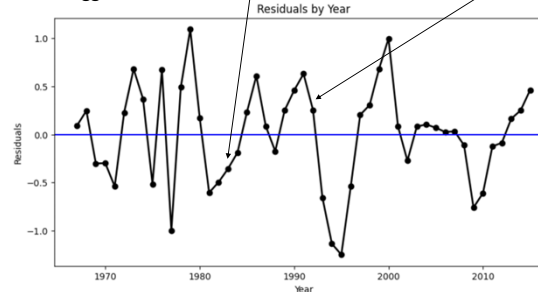
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Residuals of Coffee Demand

Amplitude appears to vary over time, so there may be heteroskedasticity.

Negative residuals from 1981-84, then positive from 1989-1992, etc.

This suggests correlated errors.



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Correcting Standard Errors – Newey West



When errors are correlated, we need to calculate standard errors of coefficients in a way that corrects for the presence of such correlation and also continue to correct for possible heteroskedasticity.

When we use “ts” in our regression command, the standard errors are corrected for possible presence of heteroskedasticity and autocorrelation.

Note that **HAC** is an acronym for “Heteroscedasticity and Autocorrelation Correction”

This correction was derived by authors Whitney Newey and Kenneth West. It is often called the Newey-West correction.

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Thinking about The Model



In general, quantity is affected by both demand and supply.

Why view the previous relationship as a demand equation?

Three features of the coffee market favor viewing this as a demand relationship.

1. The primary reason is the following: Coffee is traded in the world market. Hence, the price of coffee is determined in the world market. The US consumes only about 15% of world supply. Thus, the US is largely a price taker in the world coffee market.

Two other relevant factors are:

2. Coffee trees take years to mature. Supplier response to price in a year's time is thus likely to be quite small.
3. Fluctuations in supply occur due to weather, giving rise to year-to-year shifts in supply that are largely random.

Markets for other goods may differ, and may require a two-stage estimation approach. This would be covered in an advanced elective course.

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Interpreting Coefficients: Scaled Coefficients



We issue the command:

```
# Calculate elasticities and standardized coefficients
CoffeeReg.scaled_coefficients()
```

For interpreting a demand equation, the elasticities are of particular interest.

particular interest.

• Coffee Price Elasticity

= - .12

• Income Elasticity

= .28

• Cross-Price Elasticity

= .73

	coefs	std_coefs	elasticities
rpcofe	-0.013242	-0.191408	-0.1222
rpcinc	0.112116	0.471776	0.2821
rpcarb	0.082063	1.343097	0.7333

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Checking Functional Form: Ramsey Test



Recall that we run the Ramsey test as follows:

```
# Conduct Ramsey test
CoffeeReg.ramsey_test()
```

We obtain high p-values with one fitted term and with two fitted terms. These are favorable results for our chosen functional form.

	power	pvalue
0	2	0.771445
1	3	0.958565

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Leverage Plots

We can use the following command to view leverage plots for the intercept and the three variables:

```
# Display Leverage plots
fig = plt.figure(figsize=(12, 6))
sm.graphics.plot_partregress_grid(CoffeeReg, fig=fig)
plt.show()
```

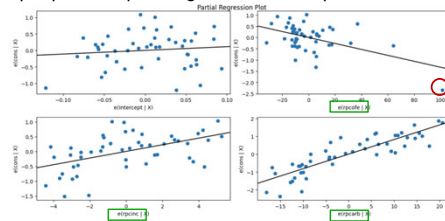
The goal of a leverage plot is to show the effect of each explanatory variable on the dependent variable while holding other explanatory variables constant.

The leverage plots nicely display the relationship of coffee demand to each of the three right-hand-side variables: coffee price (rpcofe), Per capita income (pcinc), carbonated beverage price (rpcarb).

Leverage Plot shows an "influential" observation for coffee price.

- Plot of prices earlier in the notes show that there was a year (1976) with a pronounced spike in coffee price.
- This data point is helpful from a statistical perspective in providing variation that helps us estimate consumer response to price change.

The upper-left plot can be ignored. Each other leverage plot shows the relationship of the dependent variable to a right-hand-side variable holding other variables constant. The leverage plots show no obvious cause for concern.



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Let's Investigate the Decline in Coffee Consumption

- Consumption per capita declined by 31% from 1967 to 2015 (4.6 pounds of beans per capita)
- Let t denote year, Q_t predicted consumption, I_t income, P_t coffee price, S_t soft drink price
- Income is measured in thousands.
- Our estimated equation

$$Q_t = 1.211 - 0.013P_t + 0.082S_t + 0.11I_t$$
- Change from 1967 to 2015

$$Q_{2015} - Q_{1967} = -0.013(P_{2015} - P_{1967}) + 0.082(S_{2015} - S_{1967}) + 0.11(I_{2015} - I_{1967})$$

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Why the Decline in Coffee Consumption?

- Predicted Change from 1967 to 2015

$$Q_{2015} - Q_{1967} = -0.013(P_{2015} - P_{1967}) + 0.082(S_{2015} - S_{1967}) + 0.11(I_{2015} - I_{1967})$$

	cons	rpcofe	rpcarb	pcinc
2015	10.2	86.72	66.95	38.10
1967	14.8	82.14	158.06	15.20
Difference	-4.6	4.58	-91.11	22.90

- Substituting the price and income data into equation, the predicted change in coffee consumption from 1967 to 2015 is:

$$Q_{2015} - Q_{1967} = -0.013(4.58) + 0.082(-91.11) + 0.11*22.90 = -5.01$$

The model predicts a drop of -5.01 while the actual was -4.6.

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Recap: Why the Decline in Coffee Consumption?

Predicted change from 1967 to 2015:

$$Q_{2015} - Q_{1967} = -0.013(4.58) + 0.09(-91.11) + 0.11*22.90 = -5.32$$

Our equation says predicted coffee consumption changed from 1967-2015 because:

- Coffee price increase reduced consumption by $-0.013*4.58 = -0.06$ pounds per capita
- Fall in carbonated beverage price reduced coffee consumption by $0.090*(-91.11) = -7.47$ pounds
- Rising income increased coffee consumption by $0.11*22.9 = 2.52$ pounds

The model predicts a drop of -5.01 while the actual was -4.6.

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Predictions for Years 2016 Through 2020

I was able to find data for prices of coffee and carbonated beverages through year 2020. Per capita income data are also available.

I put these additional observations into an Excel file named Coffee_Predictors.xlsx. I then read the file and calculated prediction and prediction intervals and named the resulting dataframe Pred_and_PI using the commands below:

```
# Read Coffee_Predictors.xlsx and calculate predictions and prediction intervals
Coffee_Predictors = pandas.read_excel("Coffee_Predictors.xlsx")
Pred_and_PI=CoffeeReg.prediction_intervals(Coffee_Predictors, alpha=0.05)
```

Then I plotted the data.

The resulting plot is on the next slide.

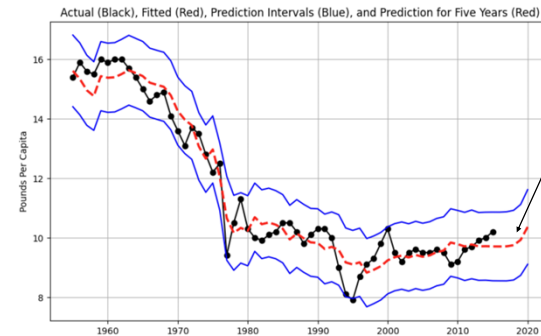
```
# Plot Actual, Predicted, and Confidence Intervals for 1955 through 2020
plt.figure(figsize=(10, 5))
plt.title("Actual(black), Predicted(red), and 95% Prediction Interval(blue)")
plt.xlabel("Year")
plt.ylabel("Actual, Predicted, and Prediction Intervals")
plt.plot(Coffee["year"], Pred_and_PI["Predicted"], color="red")
plt.plot(Coffee["year"], Pred_and_PI["Lower"], color="blue")
plt.plot(Coffee["year"], Pred_and_PI["Upper"], color="blue")
plt.scatter(Coffee_Data["year"], Coffee_Data["cons"], color="black")
plt.plot(Coffee["year"], Coffee["cons"], color="black")
```

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Prediction, Prediction Interval, and Data

Predicted values for 1967 through 2020 are included in this graph. This offers some assurance to the Coffee Growers Association that the decline in consumption has ended.



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Motivation for Analysis That Follows

For many analyses with time series data, the Newey-West (HAC) correction is all you need to do.

It is generally good practice to estimate the model to investigate the extent of correlation of the errors.

- The slides that follow show how to do that.
- In time series data, a portion of the error term in a given period may carry over to the next period.
- Allowing for correlated errors may also improve predictions.
- This is a useful form of robustness analysis

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Modeling Correlated Errors

Why might errors be correlated over time?

In the coffee demand model:

- An increase in price may result in a temporary increase in consumption from inventories.
- This unobserved consumption from inventories may give rise to correlated errors because measured consumption may differ from actual consumption if inventories are being drawn down or built up.
- Also, habit formation may cause some delay in adjusting consumption in response to changes in prices.

As another example, in an assembly line setting:

- If a problem arises that causes disruption of the assembly line on a given day, the effect may persist subsequent days until the problem is fully corrected.

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Modeling Correlated Errors

The simplest and most common form of correlated errors is the following.
Each period, a fraction of the error from the previous period is carried forward, and a new random error is added.

Example: Suppose the following:

- The error term in year t-1 for our coffee demand model is .3, i.e., coffee demand last year was higher than predicted by .3 pounds per capita.
- One half of that error, $.5 \times .3 = .15$, is carried over to year t due, say, to habit formation.
- Some new random event affects consumption by amount .11 in year t.
- The error term in year t is then the carryover from last period plus the new random element: $.5 \times .3 + .11 = .15 + .11 = .26$

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Modeling Correlated Errors

In symbols, let

u_{t-1} be the error term in our regression last period,

ρ_1 the fraction carried forward to period t, and

ε_t a new random event in period t.

Then the error term in period t is: $u_t = \rho_1 u_{t-1} + \varepsilon_t$

- This is a model with “first-order serial correlation” of the error terms. This is also called first-order autocorrelation or, alternatively, a first-order autoregressive model, often denoted by the shorthand AR(1).
- The first-order correlation coefficient is ρ_1 .

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Estimating Models with Correlated Errors

Suppose you have a model and you want to estimate the model to investigate whether the errors are correlated.

The commands to estimate the model and view the results are presented in the next slide.

The coefficients reported are the usual estimates of the unknown population β 's in the regression model.

The ar1 coefficient is an estimate of ρ_1 in the equation for the error terms:

$$u_t = \rho_1 u_{t-1} + \varepsilon_t$$

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Correlated Errors in Coffee Demand Model

We estimate and view results with the following commands. I named my model `cof_arima`

```
# Estimate with first-order autoregressive errors.
# First-order autoregressive the default in the tprstats arima implementation
cof_arima = tprstats.model("arima", "cons=rpcofe+rpcinc+rpcarb", Coffee_Data)
cof_arima.summary()
```

- The estimate of the correlation coefficient is .435 and is significant.
- The rest of coefficients are interpreted as before.
- The coefficients and standard errors are changed somewhat relative to our previous estimates.

	coef	std err	z	P> z	[0.025	0.975]
const	2.4158	2.367	1.021	0.307	-2.223	7.054
rpcofe	-0.0172	0.003	-6.756	0.000	-0.022	-0.012
rpcinc	0.0948	0.049	1.930	0.054	-0.001	0.191
rpcarb	0.0773	0.012	6.486	0.000	0.054	0.101
ar.L1	0.4349	0.117	3.702	0.000	0.205	0.665
sigma2	0.2194	0.051	4.300	0.000	0.119	0.319

$\hat{\rho}_1$

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Comparing the Estimates

- Coefficient estimates are compared below.
- We see that the coefficients are broadly similar.**
- Allowing for correlated errors somewhat increased the absolute value of the coefficient of rpcofe and somewhat decreased the coefficients of the other two variables.
- For example, the estimated effect of income is somewhat lower.
- This provides useful sensitivity analysis.

Without AR(1)			With AR(1)		
	Coefficients	p-values		Coefficients	p-values
Intercept	1.121	0.566	const	2.416	0.307
rpcofe	-0.013	0.001	rpcofe	-0.017	0.000
rpcinc	0.112	0.002	rpcinc	0.095	0.054
rpcarb	0.082	0.000	rpcarb	0.077	0.000
			ar.L1	0.435	0.000

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Higher Order Correlation

First-order correlation, as in our coffee model, is very common. Sometimes higher order correlation of the errors may occur.

- We write this as: $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \varepsilon_t$

I estimated the model with both first-order and second-order autocorrelation.

The estimated second-order correlation coefficient was -.17 with p-value .21.

Hence, we can conclude second-order autocorrelation is not a concern in our coffee demand model.

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Implications of Our Analysis

Our estimated model provides information for the Coffee Grower's Association about the factors that affect coffee demand.

Our model suggests that the large decrease in the price of carbonated beverages decreased the demand for coffee.

From the graph of prices that we presented at the [beginning of](#) these slides, we see that the price of carbonated beverages is declining much less rapidly in recent years.

This coupled with continued rise in income suggests that the decline in coffee demand may have largely run its course.

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Model with Logarithms of Variables

I also estimated the model with logarithms of variables. The coefficients are all significant at the 5% level and all have the anticipated algebraic signs.

As you know, the coefficients are elasticities when both dependent and independent variables are in log form. This makes the coefficients easy to interpret.

The Ramsey test also gives relatively high p-values for this model.

It is somewhat unusual for models without logs and with logs to fit well, which is an interesting feature of this application.

OLS Regression Results							power	pvalue
Dep. Variable:	log(const)	R-squared:	0.867					
Model:	OLS	Adj. R-squared:	0.859					
No. Observations:	49	F-statistic:	117.0				2	0.377992
Covariance Type:	HAC	Prob (F-statistic):	2.89e-21				3	0.672344
	coef	std err	t	P> t	[0.025	0.975]		
Intercept	-3.1879	1.516	-2.102	0.041	-6.242	-0.134		
log(rpcofe)	-0.1642	0.045	-3.686	0.001	-0.254	-0.074		
log(rpcinc)	0.5339	0.183	2.922	0.005	0.166	0.902		
log(rpcarb)	1.0098	0.185	5.451	0.000	0.637	1.383		

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Further Thoughts on Demand Estimation



We previously studied the demand for natural gas.
 You might be wondering whether it is valid to estimate a demand function for natural gas without taking account of supply response.
 The answer is that natural gas supply does respond to price changes, but there is little supply change in a given year to a price change that year.
 Changes in supply happen over the years as new gas wells are drilled and existing wells are depleted.
 A price increase in a given year does not have a substantial impact on supply that year, but will impact supply in future years.
 Hence, in a given year, price adjusts demand to the fixed supply that year.
 In other markets, supply and demand may change within a given year. Estimation methods are available for such markets, but time does not permit us to develop those methods in our course.

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Something To Keep In Mind...



- Suppose you estimate a model and you obtain an ar1 coefficient close to one or you obtain a coefficient of a lagged dependent variable close to one.
- In such circumstances, it may be appropriate to estimate the model with all variables expressed in period-to-period changes rather than levels.
 - We will now investigate such an application.

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Credit Card Interest Rates Reconsidered



In Assignment 1, you estimated the regression below with change in credit card interest rates (dCredcrd) as the dependent variable and change in treasury interest rates (dTreas) as the independent variable. We obtained an estimated coefficient of .558.

```
=====
Dep. Variable:          dCredcrd  R-squared:                0.288
Model:                  OLS       Adj. R-squared:           0.281
No. Observations:      97        F-statistic:              6.102
Covariance Type:       HAC       Prob (F-statistic):       0.0153
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----+-----
Intercept    0.0631      0.046      1.363    0.176    -0.029    0.155
dTreas       0.5582      0.226      2.470    0.015     0.110    1.007
=====
```

Why not estimate a model with credit card interest rates (Credcrd) as the dependent variable and treasury interest rates (Treas) as the independent variable? Let's investigate.

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Estimate Model of Credcrd versus Treas



- From the regression below, we see that the coefficient of Treas is .9766 which is close to 1. From this regression, we would have concluded that the credit card interest rates follow very closely treasury rates. Why the difference in the result below relative to the .558 estimate we obtained just above? The answer is that the error terms in the regression below are highly correlated with each other, as demonstrated next.

```
=====
Dep. Variable:          Credcrd  R-squared:                0.390
Model:                  OLS       Adj. R-squared:           0.384
No. Observations:      98        F-statistic:              10.29
Covariance Type:       HAC       Prob (F-statistic):       0.00182
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----+-----
Intercept    12.4387      0.337     36.878    0.000     11.769    13.108
Treas        0.9766      0.305      3.207    0.002      0.372     1.581
=====
```

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Estimate Model of Credcrd versus Treas with AR(1)



Estimating the model allowing for first-order autocorrelation of the errors, we obtain an estimated correlation coefficient of .992.

The coefficient of Treas is .562 which is very close to the .558 we obtained using dCredcrd vs dTreas. However, the standard error is .048, much smaller than the value of .226 we obtained with dCredcrd vs dTreas. The standard error below is biased.

```
=====
Dep. Variable:      Credcrd   No. Observations:      98
Model:             ARIMA(1, 0, 0)   Log Likelihood          -47.821
Date:              Wed, 11 Mar 2026   AIC                     103.642
Time:              17:20:31          BIC                     113.982
Sample:            0              HQIC                      107.825
Covariance Type:    opg
=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
const         15.0138      3.380      4.441    0.000      8.388     21.639
Treas          0.5620      0.048     11.753    0.000      0.468     0.656
ar.L1          0.9921      0.015     66.802    0.000      0.963     1.021
sigma2         0.1489      0.011     13.018    0.000      0.127     0.171
=====
```

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Message from The Credit Card Analysis



Some terminology: Variables Credcrd and Treas are in “levels”. Variables dCredcrd and dTreas are “changes of variables”. Also called first differences. If you have an estimated model with autocorrelation coefficient that is relatively high, try estimating the model with changes in the variables as we did with dCredcrd versus dTreas. This is likely to give you more valid results?

It is natural to ask how high an autocorrelation coefficient should be to prompt you to estimate with changes rather than levels. This varies with applications, but an autocorrelation coefficient of .80 or higher would point toward estimation using changes rather than levels of variables.

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Summer Internship



A former Tepper MBA student had a summer internship with for a US company that exported goods overseas. Let's name this company AMZ. Suppose people in country H purchase goods from AMZ.

Suppose the exchange rate between the US and country H changes with the dollar rising relative to the foreign currency. This makes the goods from AMZ more expensive in country H. Conversely, if the exchange rate between the US and country H declines relative to the foreign currency, goods from AMZ become expensive in country H.

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Summer Internship Continued



She determined that the appropriate regression for each exported product was to use the change in logarithm of sales of the product as the dependent variable and the changes in the logarithm of the exchange rate as the independent variable.

Her findings helped the company in pricing goods to various countries.

She did such an analysis for many goods exported by AMZ. The company was very impressed by her work and recruited her for a fulltime position when she graduated from Tepper.

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Summary

We analyzed data that vary over time studying a model of coffee demand.

We discussed why errors may be correlated over time.

We saw that both heteroskedasticity and autocorrelation may be present, and we saw how to correct standard errors using the Newey-West correction.

We also saw how to estimate models including estimation of the correlation of the error terms.

We saw that when a correlation coefficient that is relatively high, it may be appropriate to estimate in changes rather than levels.

We found this to be the case in analyzing the response of credit card rates to changes in the Treasury rate.

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